Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling

1 Introduction

Layered van der Waals solids are often prepared as thin films, where their anisotropic crystal structures make crystallite orientation and phase central to the measured response.[1–3] Depending on the substrate and growth conditions, the crystallites may align their layer normals with the film normal while retaining little common in-plane azimuthal registry.[3] We refer to this uniaxially textured limit as a two-dimensional (2D) powder: individual crystallites remain three-dimensional, but one crystallographic axis is preferentially aligned while their rotations about it are approximately random.[4, 5] The resulting reciprocal-space distribution produces characteristic specular and off-specular features that an area detector records simultaneously, providing access to their detector-space positions, apparent line shapes, azimuthal extents, and intensities for corrected quantitative comparison.[6–8]

The difficulty is that the same detector features can arise from several correlated experimental and sample-dependent causes, including finite wavelength bandwidth, beam divergence, grazing-incidence refraction, sample misalignment, mosaic spread, and disorder.[4, 8–10] One solution is a forward model that starts from a crystallographic and structural hypothesis and propagates it through the incident beam distribution, sample geometry, scattering kinematics, calibrated detector mapping, and finite detector integration. Keeping the calculation in detector space allows the assumptions that generate each feature to be tested against the measured two-dimensional intensity distribution rather than only a one-dimensional reduction. Figure 1 places the 2D-powder state between the single-crystal and 3D-powder limits in real, reciprocal, and detector space.

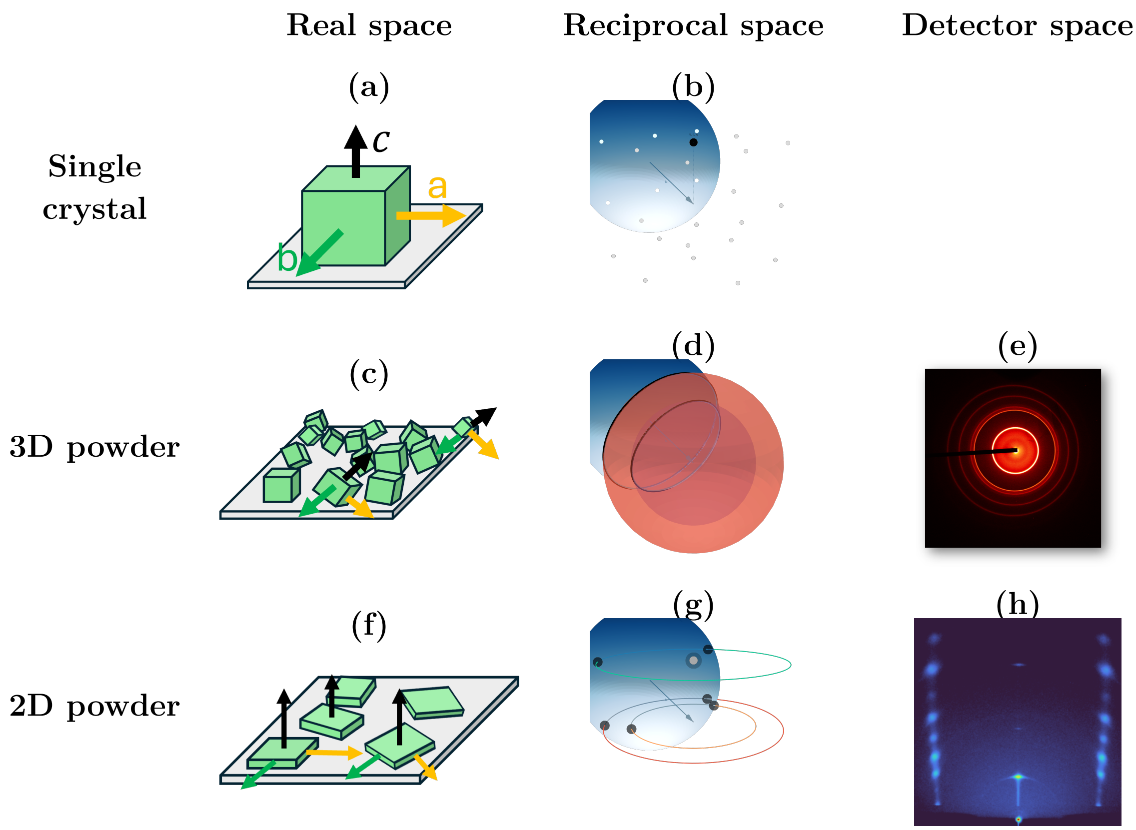

Figure 1: Diffraction across three orientational limits in real, reciprocal, and detector space. In the reciprocal-space panels (b,d,g), the blue Ewald sphere has radius $|\mathbf{k}_i| = 2\pi/\lambda$. Dark points or curves mark reciprocal-space intensity intersected by that sphere. (a,b) A single crystal has isolated reciprocal-lattice points, so isolated crossings give sparse detector spots. (c–e) Random 3D orientations spread a reflection over a spherical shell, whose intersection with the Ewald sphere maps to a Debye–Scherrer ring. (f–h) A 2D powder retains a common film normal but is random in in-plane azimuth. Off-specular reflections become rings about the film-normal reciprocal-space axis. Their Ewald-sphere crossings map to paired detector features when both fall within detector acceptance. The $00L$ intensity remains near the specular direction.

Figure 1 summarizes the real-space, reciprocal-space, and detector-space consequences of these three orientational limits. Here $\mathbf{k}_i$ and $\mathbf{k}_f$ are the incident and exit wavevectors, and $\mathbf{G}$ is a reciprocal-lattice vector. With this convention, the Ewald sphere is centered at $-\mathbf{k}_i$, passes through the reciprocal-space origin, and has radius $|\mathbf{k}_i| = 2\pi/\lambda$. Elastic diffraction occurs where reciprocal-space intensity satisfies $|\mathbf{k}_i + \mathbf{G}| = |\mathbf{k}_i|$. Each allowed point defines an exit direction $\mathbf{k}_f = \mathbf{k}_i + \mathbf{G}$.[11] For a single crystal, reciprocal space consists of isolated Bragg points [Fig. 1(b)]. Only isolated sphere–point crossings produce detector spots, so the detector pattern is sparse. For a 3D powder, random crystallite orientations spread each reflection over a spherical shell [Fig. 1(d)]. The sphere–shell intersection is a circle that maps to the Debye–Scherrer ring in Fig. 1(e).[11] For a 2D powder, azimuthal disorder about the common c axis spreads each off-specular Bragg point into a ring about the film-normal reciprocal-space axis, while specular ($00L$) intensity remains near that axis [Fig. 1(g)].[4, 6] The Ewald sphere generally cuts an off-specular ring at two points, which map to paired detector features when both directions fall within detector acceptance [Fig. 1(h)].[6, 7, 12] Mosaicity then broadens these ideal intersections into arcs and cap-like intensity rather than changing the basic diffraction condition.[4, 6]

Figure 2 shows a representative measured 2D-powder pattern in which the Bragg peaks, arcs, and streaks retain this anisotropic detector-space intensity distribution.

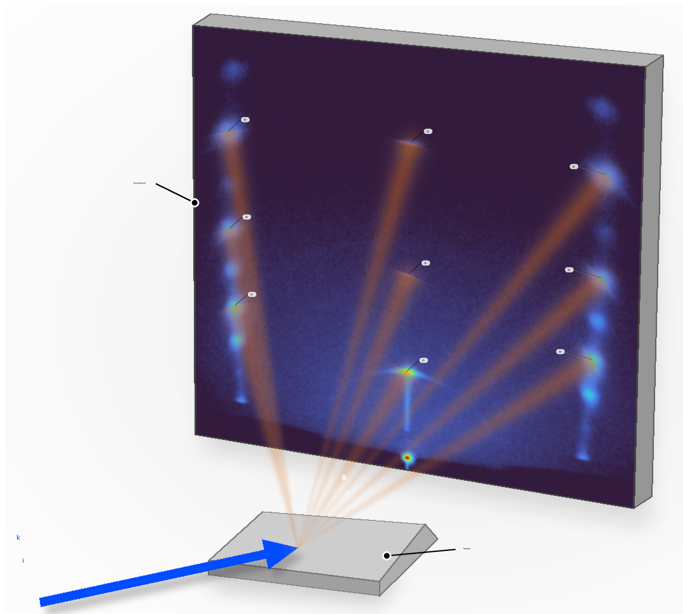

Figure 2: Illustration of a 2D-powder WAXS measurement using a measured PbI_2 sample. The incident beam strikes the film, and diffracted intensity is recorded directly on the area detector. Bright features indicate Bragg peaks, while the streaks between them and arcs illustrate how oriented crystallites, finite mosaic spread, and detector geometry distribute reciprocal-space intensity across the detector image.

This hybrid detector-space information is what makes the 2D-powder problem both rich and difficult. Many standard WAXS workflows reduce area-detector images to 1D profiles by azimuthal integration, often with masking near the direct beam and specular region.[7, 8, 13–15] Those steps suit nearly isotropic powders, but for 2D powders they deweight or remove off-specular arcs and specular caps that constrain out-of-plane correlations, stacking, and mosaic tails.[6, 7] Stacking disorder is also commonly inferred from 1D interference models and refinements,[16–18] which do not directly exploit the two-dimensional anisotropy of oriented-film scattering. More general intermediate orientation distributions can be treated with orientation-distribution-function (ODF) refinements or preferred-orientation corrections,[19–24] but those approaches are more general than needed for the narrowly constrained 2D-powder case considered here.[23, 24] Recent radial-profile approaches have made thin-film GIXD substantially more quantitative, but peak overlap and background separation can remain limiting in strongly anisotropic patterns, especially at small incidence angles.[25] For the layered films considered here, the measured morphology already motivates axial symmetry about the film normal and a low-dimensional out-of-plane mosaicity kernel.[4, 5, 26] We therefore assume this physically motivated 2D-powder orientation distribution a priori and refine it with a detector-space intensity forward model, using off-specular arcs and specular-family line shapes as direct constraints.[6, 7]

Prior work has established the main ingredients needed for such a framework, including grazing-incidence scattering formalisms,[27, 28] reciprocal-space mapping and indexing for oriented thin films,[6–8, 12] detector-space visualization and indexing of GIXS/GIWAXS patterns,[29] and forward calculations of diffraction from textured films.[9] Most directly, Smilgies and Li calculated grazing-incidence diffraction-spot positions in detector space rather than nonlinearly transforming the full detector image into reciprocal space.[29] The present work extends that geometric starting point to a continuous detector-coordinate intensity calculation that combines the incident beam phase space, calibrated sample and detector geometry, a continuous crystallite-orientation distribution, structure-factor weighting, and finite detector integration. Standard planar-interface refraction and attenuation corrections are included where required, but they are not a principal contribution of this work.

Here we present the resulting detector-space intensity forward model for WAXS from 2D-powder films and apply the same framework to ordered Bi_2Se_3 and Bi_2Te_3 films.[2, 3] Figures 7 and 8 show the $\theta_i = 5°$ condition, where calculated hexagonal rod-family trajectories track selected measured Bragg-feature locations and the corresponding projected profiles reproduce the principal peak positions and apparent line shapes. Images collected at 5°, 10°, and 15° entered the staged refinement. The displayed 5° comparisons therefore show agreement for a condition used in that refinement rather than an independent incidence-angle prediction. The result is a direct detector-space comparison that retains anisotropic information obscured by a conventional one-dimensional reduction.

The paper first develops the diffraction geometry and detector-space intensity calculation, then uses the measured low-L features and their reciprocal-space geometry to motivate the mosaicity model. The incident beam phase space and wavelength dependence are introduced as inputs to the same calculation, while standard planar-interface corrections and the near-critical specular diagnostic are documented chiefly in the Supporting Information. The ordered Bi_2Se_3 and Bi_2Te_3 test cases are followed by the staged refinement workflow used to construct those comparisons. PbI_2 is treated afterward as a separate stacking-disorder extension that retains the detector-space projection while changing the structure factor along fixed in-plane-momentum trajectories. The discussion closes by relating the intensity calculation to detector-space indexing and structural interpretation.

2 Diffraction Model

The detector-space intensity model begins from the same reciprocal-space intersections used for detector-space indexing.[29] The calculation then follows one physical sequence. A weighted incident state specifies beam position, propagation direction, and wavelength. The ray is intersected with the sample geometry, standard planar-interface corrections determine the internal propagation state, and the wavelength-specific Ewald condition restricts the continuous reciprocal-space intensity generated by the crystallite-orientation distribution. Each accepted exit direction is returned to the external medium, mapped to the calibrated detector plane, and integrated over the same finite detector regions used for the measured profiles. The detailed coordinate transforms, ray-plane intersections, and integration formulas are given in the Supporting Information.

2.1 Diffraction Geometry

We first consider a single incident state and a perfectly aligned crystallite, with beam divergence, refraction, and absorption suppressed. Let $\mathbf{k}_i$ denote the incident wavevector and $\mathbf{k}_f$ the scattered wavevector. For x-ray wavelength λ, elastic scattering preserves their magnitudes,

$$|\mathbf{k}_i| = |\mathbf{k}_f| = \frac{2\pi}{\lambda}. \tag{1}$$

Equation (1) sets the Ewald-sphere radius. The momentum transfer is

$$\mathbf{Q} = \mathbf{k}_f - \mathbf{k}_i. \tag{2}$$

For reflection (h, k, l), $\mathbf{G}_{hkl}$ is the corresponding reciprocal-lattice vector. Diffraction occurs when

$$\mathbf{Q} = \mathbf{G}_{hkl}, \qquad \mathbf{G}_{hkl} = h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^*, \tag{3}$$

where $\mathbf{a}^*$, $\mathbf{b}^*$, and $\mathbf{c}^*$ are reciprocal-lattice basis vectors. Combining Eqs. (2) and (3) gives

$$\mathbf{k}_f = \mathbf{k}_i + \mathbf{G}_{hkl}. \tag{4}$$

For fixed $|\mathbf{G}_{hkl}|$, the possible orientations of that vector form a Bragg sphere about the reciprocal-space origin. The elastic condition selects the intersections of this locus with the Ewald sphere.

The incident beam is represented by a finite set of weighted source states in position, direction, and wavelength. A source ray is first intersected with the sample plane. The accepted intersection fixes the scattering origin, the illuminated footprint, and the local incidence angle. Finite sample support can be imposed when its dimensions are known. Otherwise, the film is treated as an unbounded plane in its in-plane directions. The footprint is therefore the distribution of accepted ray intersections, not an additional displacement applied later.

The detector provides a second geometric acceptance condition. An allowed exit ray is propagated from the sample intersection to the calibrated detector plane. Detector distance, beam-center coordinates, detector tilt, pixel pitch, and active area determine whether the ray is recorded and where it appears. This step is performed in continuous detector coordinates before intensity is integrated over a native pixel or a declared finite angular bin. Figure 3 shows the sample-to-detector mapping, while Fig. 4 shows the sample position and alignment variables.

Reflection construction is deterministic. Integer (h, k) rod identities are enumerated over the detector-accessible reciprocal-space range and systematic absences are imposed by the selected structural model. The incident source distribution is represented by weighted numerical quadrature. The crystallite-orientation distribution remains a continuous density, and its Ewald intersections are solved for each retained rod rather than generated by a separate cloud of stochastic crystallite events. The resulting detector-coordinate density is then summed over source states.

A reflection label in a detector panel identifies a physical (h, k, l) member or the m-indexed in-plane family defined below. A plus or minus sign distinguishes the two detector-side intersections of one off-specular family. For a selected profile, the calculated trajectory defines a finite integration region and its reciprocal-space coordinate. Measured signal and calculated detector density are evaluated with the same calibrated coordinate map, integration limits, and normalization. Within each profile bin, signal and detector-area normalization are accumulated before their ratio is formed. The operation is therefore a matched finite projection, not a separate peak-identification procedure.

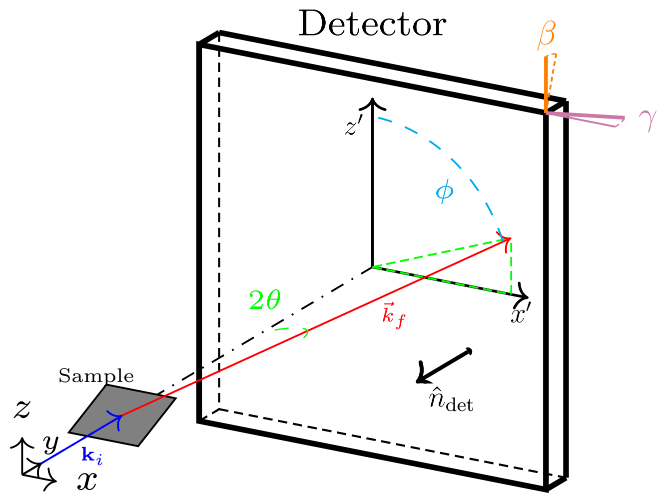

Figure 3: Overall laboratory geometry. The incident wavevector $\mathbf{k}_i$ strikes the sample, and a representative exit wavevector $\mathbf{k}_f$ reaches the calibrated detector at scattering angle 2θ and detector-plane azimuth ϕ. The detector axes x' and z', detector normal $\hat{\mathbf{n}}_{\mathrm{det}}$, distance D, and tilts β and γ determine the map from the exit direction to continuous detector coordinates. The ray-plane and pixel-coordinate equations are given in the Supporting Information.

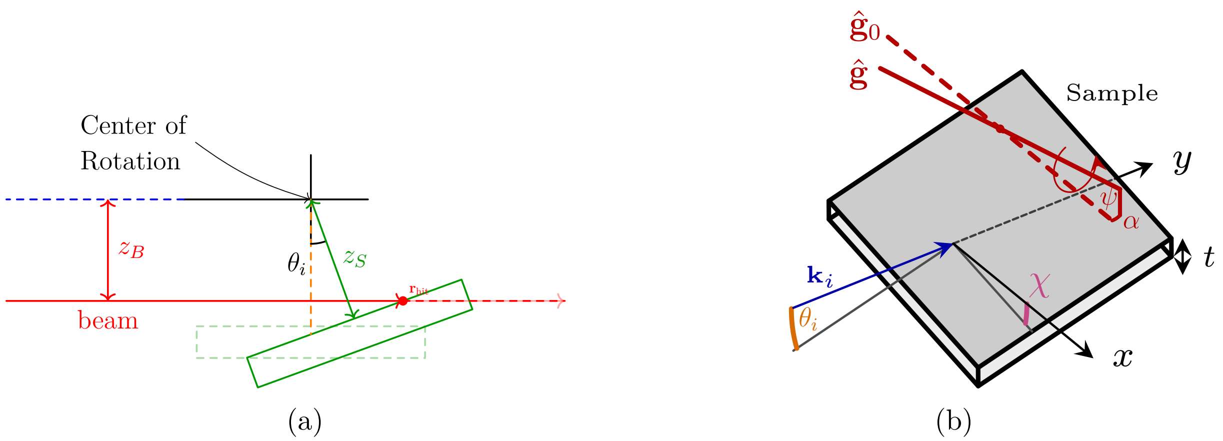

Figure 4: Sample position and alignment geometry. (a) The red incident ray intersects the film at $\mathbf{r}_{\mathrm{hit}}$. The quantities z_B and z_S are the beam and sample-surface offsets from the goniometer center, and θ_i is the incidence angle. Dashed outlines show the zero-angle reference. (b) The incident wavevector is $\mathbf{k}_i$, x and y are laboratory axes, t is the film thickness, and χ and θ_i describe sample rotations about y and x, respectively. The dashed $\hat{\mathbf{g}}_0$ and solid $\hat{\mathbf{g}}$ are the nominal and misaligned goniometer-axis directions. The angles ψ and α rotate that direction about laboratory y and then x, respectively.

2.2 Structure Factor

The reciprocal basis in Eq. (3) includes the factor 2π. For an atom a with fractional coordinates (x_a, y_a, z_a), occupancy o_a, and atomic form factor $f_a(Q)$, the conventional crystallographic structure factor is[30]

$$F_{hkl} = \sum_a o_a f_a(Q)\, T_a(hkl)\, \exp[2\pi i(hx_a + ky_a + lz_a)]\,. \tag{5}$$

Here $Q = |\mathbf{Q}|$, and $T_a(hkl)$ is the crystallographic displacement factor constructed from the reported isotropic or anisotropic U or B convention used by the selected structural model. If form-factor tables use $s = \sin\theta/\lambda$, then $s = Q/(4\pi)$. The intensity weight of an allowed member is proportional to $|F_{hkl}|^2$. The Supporting Information states how the current calculation handles isotropic CIF displacement values and the shared uniaxial film-frame damping used by the ordered-film parameterization.

For the film-normal unit vector $\hat{\mathbf{n}}$, the projected coordinates used for detector-space profiles are

$$Q_z = \mathbf{Q} \cdot \hat{\mathbf{n}}, \qquad \mathbf{Q}_\parallel = \mathbf{Q} - Q_z\hat{\mathbf{n}}, \qquad Q_R = |\mathbf{Q}_\parallel|, \qquad L = \frac{Q_z c}{2\pi}. \tag{6}$$

For an ideally aligned (h, k, l) Bragg vector, $L = l$. Away from an exact Bragg point, L is a continuous projected coordinate rather than a Miller index.

For the hexagonal basal plane, integer in-plane indices (h, k) are assigned the radial family label

$$m = h^2 + hk + k^2, \tag{7}$$

with in-plane reciprocal-space radius

$$Q_R = \frac{4\pi}{\sqrt{3}\,a}\sqrt{m}. \tag{8}$$

Reflections with the same m share the same ideal in-plane radius and are grouped under one radial-family label for figures and profile selection. The calculation still evaluates every allowed (h, k, l) member separately before summing accepted contributions. The label m is therefore not a crystallographic multiplicity, and no second multiplicity factor is applied after those members have been summed.

2.3 Mosaicity

Mosaicity is the out-of-plane orientational spread of the layered crystallites about the mean film normal illustrated in Fig. 1(f–h). It is distinct from the random in-plane rotation that defines the two-dimensional powder limit. Random in-plane rotation generates an m-indexed rod family at a common in-plane reciprocal-space radius Q_R, whereas mosaicity spreads each member over crystallite tilt. On an area detector, those two operations appear differently.

Random azimuthal rotation generates paired off-specular arcs for $m \neq 0$. Out-of-plane tilt broadens those arcs and distributes $m = 0$ intensity away from the ideal specular direction.

Figure 5 shows a Bi_2Se_3 detector image at $\theta_i = 5°$ with the low-L, $m = 0$ region enlarged. The inset labels features at the nominal 003 and 006 locations in the displayed detector coordinate. The 003 region forms a visibly extended streak, while the 006 region is more localized. Their apparent relative intensity and the length of the 003 streak cannot be inferred from the displayed angular positions alone. A quantitative explanation requires one common intensity scale together with the measured wavelength distribution, structure factors, refined alignment, orientation distribution, background treatment, and detector mapping. Figure 5 is therefore used as the observation that motivates testing a broad orientation component, not as a common-scale relative-intensity result or as a completed assignment of the streak mechanism.

The bright near-origin intensity lies in the near-critical specular regime. It is excluded from the mosaic argument and treated as a separate standard reflectivity diagnostic in Sec. 2.5 and the Supporting Information. Bandwidth, divergence, alignment, background, and mosaicity remain coupled in the breadth and visibility of the higher-L features.

Measured low-L features of Bi_2Se_3

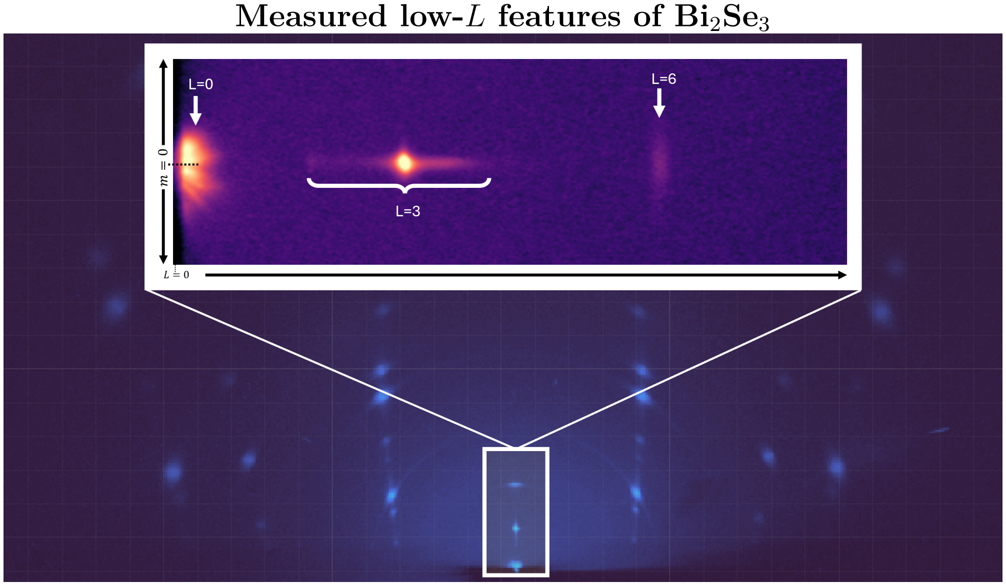

Figure 5: Measured low-L, $m = 0$ detector region from the 5° Bi_2Se_3 image. The bright near-origin intensity lies in the near-critical specular regime and is excluded from the mosaic-tail inference. The inset marks the nominal 003 and 006 locations and shows the extended streak through the 003 region. The feature positions alone do not determine their relative excitation or the origin of the streak. Those questions require a common-scale calculation using the measured wavelength distribution, structure factors, refined geometry, orientation distribution, and detector mapping. The figure therefore motivates testing a broad orientation component without claiming that the present image establishes its necessity.

The off-specular rod-family arcs and the off-condition $m = 0$ features sample different angular ranges of the same orientation distribution. Strong off-specular arcs predominantly constrain crystallites close to the mean film normal. Weak $m = 0$ intensity away from the nominal specular condition is more sensitive to the low-probability population at larger tilt. We represent those two angular scales with the compact phenomenological density

$$\omega_{\mathrm{w}}(\Delta\theta; p_{\mathrm{tail}}, \Gamma, \sigma) = p_{\mathrm{tail}} L_{\mathrm{w}}(\Delta\theta; \Gamma) + (1 - p_{\mathrm{tail}}) G_{\mathrm{w}}(\Delta\theta; \sigma), \qquad 0 \le p_{\mathrm{tail}} \le 1. \tag{9}$$

Here G_{w} is the Gaussian-like narrow core, L_{w} is the Lorentzian-like broad component, p_{tail} is its weight, and $\Delta\theta$ is crystallite tilt relative to the mean layer normal. The two widths are independent and share one center. The Lorentzian-like form is a convenient broad-tailed parameterization, not a claim that the physical distribution is uniquely Lorentzian. The present overlays show that this two-component form can represent the selected profiles. Establishing necessity requires both a fixed-parameter no-tail counterfactual and a fairly re-optimized Gaussian-only comparison.

Figure 6 connects the two parts of Eq. (9) to reciprocal-space observables. In panels (a) and (b), the compact purple segment marks the narrow core and the dashed orange extension marks the low-probability broad component. For $m \neq 0$, the strong detector arcs primarily sample the compact region near the nominal reciprocal vector. For $m = 0$, intensity farther from the film-normal axis requires larger crystallite tilts and is therefore more sensitive to the broad component. The same angular density generates both effects. The radial coordinate is folded onto $Q_R \ge 0$, so opposite signed in-plane tilt components map to the same radial locus.

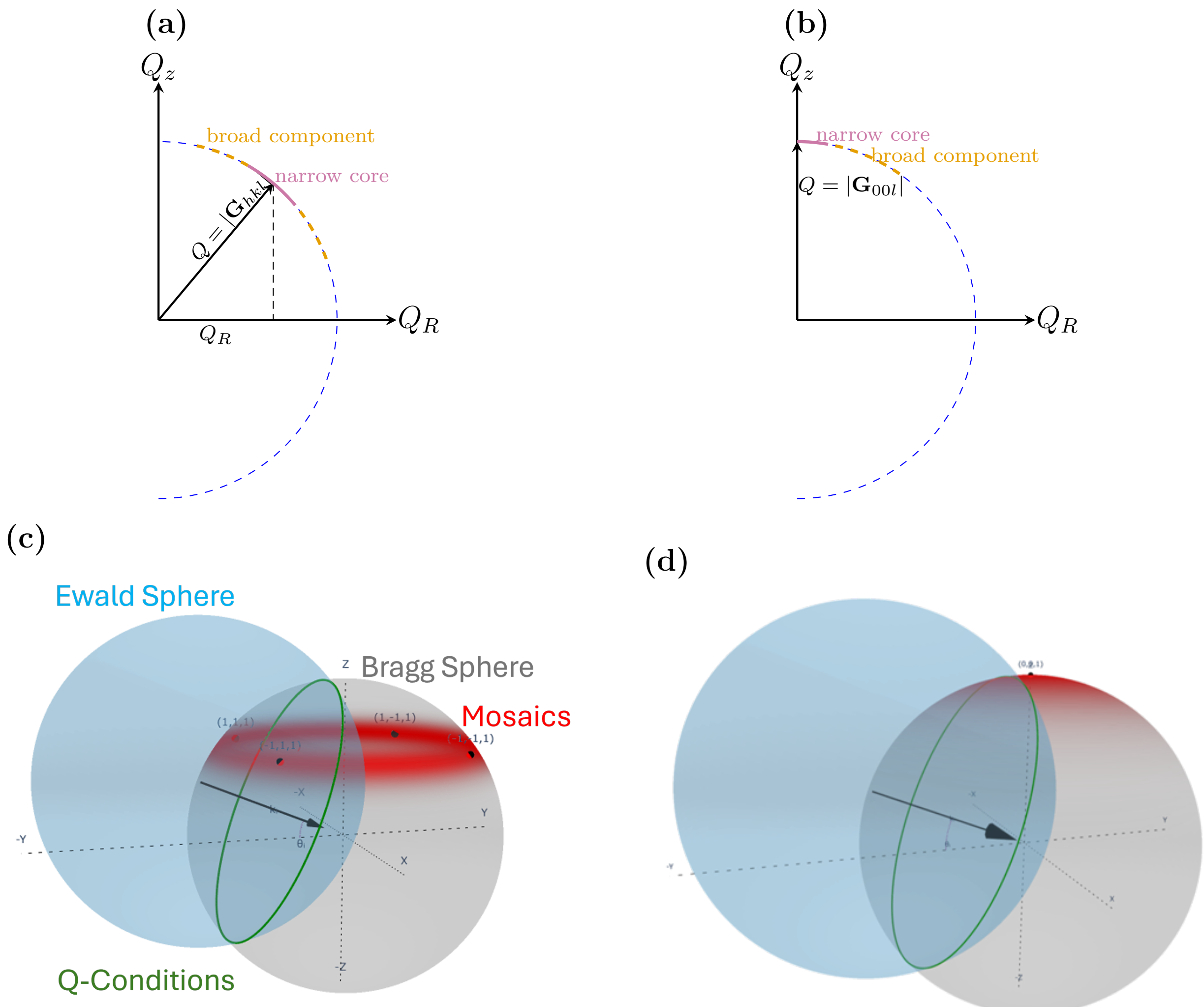

Figure 6: How the two angular scales in Eq. (9) enter the detector calculation. (a) For $m \neq 0$, the reciprocal vector has radial and film-normal components Q_R and Q_z. The purple segment identifies the narrow core that predominantly controls the strong off-specular arc width. The dashed orange extension identifies the broad, low-probability component. (b) For $m = 0$, the same distribution begins on the film-normal axis. Intensity at larger Q_R samples larger crystallite tilts and is therefore more sensitive to the broad component. Opposite signed tilts fold onto the same $Q_R \geq 0$ locus. (c,d) Three-dimensional views place the corresponding orientation distributions relative to the Ewald-sphere intersection. The schematics explain the geometric sensitivity of the two feature classes. They are not detector simulations and do not replace the no-tail and Gaussian-only controlled comparisons required to establish necessity.

2.4 Incident-beam phase space and wavelength

The incident beam is represented by weighted states in position, propagation direction, and wavelength. Position and direction determine where the ray reaches the film and its local incidence condition. Wavelength fixes the magnitude of the incident wavevector and therefore the Ewald-sphere construction used for that state. These variables are introduced together because they are inputs to one forward calculation rather than independent corrections applied to a completed detector image.

For wavelength state λ_j with spectral weight w_j,

$$k_0(\lambda_j) = \frac{2\pi}{\lambda_j}, \qquad \mathbf{k}_i(\lambda_j) = k_0(\lambda_j)\hat{\mathbf{s}}_{i,j}, \qquad |\mathbf{k}_i(\lambda_j) + \mathbf{G}| = k_0(\lambda_j). \tag{10}$$

With the reciprocal-space origin fixed, the Ewald sphere for state j is centered at $-\mathbf{k}_i(\lambda_j)$ and has radius $k_0(\lambda_j)$.[11] Changing wavelength therefore changes both the center and the radius. The full vector condition in Eq. (10) is evaluated for each weighted direction and wavelength state, and the resulting detector intensities are summed incoherently. The source-state construction and convergence checks are given in the Supporting Information.

Standard planar-interface refraction and attenuation corrections are applied to the incident and exit fields before the accepted contribution is mapped to the detector. They account for the change between external and internal propagation directions, transmission at the film boundary, and the finite depth sampled by the incident and exit fields. These corrections are included for physical completeness but are not a principal result of this work. Their boundary-condition equations, branch convention, attenuation integral, and wavelength dependence are given in the Supporting Information.

2.5 Near-critical specular response

The bright near-origin $m = 0$ intensity in Fig. 5 lies in the near-critical specular regime. Its physical origin is distinct from the orientational broadening used to describe the higher-L $m = 0$ features. Refraction, evanescence, absorption, roughness, and multiple reflection at the film interfaces can control this region, for which a standard Parratt recursion is the appropriate optical diagnostic.[31]

The near-critical region is therefore excluded from evidence for the broad mosaic component. The archived reflectivity comparison is not retained as a main-text quantitative result because its measured and complete calculated curves are not available as direct matched arrays. The Supporting Information records the Parratt calculation and the empirical crossover diagnostic used in the earlier analysis, together with this evidence limitation. No central conclusion of the present manuscript depends on that crossover construction.

3 Test cases: Bi_2Se_3 and Bi_2Te_3 films

Ordered Bi_2Se_3 and Bi_2Te_3 films provide test cases for the detector-space intensity calculation. Detector images collected at incident angles of 5°, 10°, and 15° entered the staged refinement. Figures 7 and 8 show the 5° condition because it displays both the selected off-specular rod families and the $m = 0$ features clearly within one detector image. The displayed comparisons therefore show agreement for a condition included in the refinement rather than an independent incidence-angle prediction.

For each selected family, the calculated trajectory defines the center of a finite integration region and the corresponding Q_z or L coordinate. The measured detector signal and the calculated detector-coordinate density are transformed with the same calibrated coordinate map. Within each profile bin, signal and detector-area normalization are accumulated over the same finite region before division. The plotted curves therefore compare the same detector-derived observable rather than an ideal reciprocal-space line cut.

In the detector panels, solid colored curves mark calculated trajectory centerlines. Semi-transparent bands show the finite regions used for the profiles, and dashed outlines mark their boundaries. The central blue region follows the $m = 0$ family. Paired off-specular regions are the two detector-side intersections of one m family, identified by plus and minus signs. In the profile panels, black solid curves are measured data and orange dashed curves are the corresponding calculations. The horizontal coordinate is L, with integer markers denoting nominal Bragg conditions along the selected trajectory. Off-specular rows are plotted on a linear scale, while the $m = 0$ row is plotted logarithmically to display weak off-condition intensity. The amplitudes in these panels are not interpreted as one common physical scale across different families.

Mosaicity, beam divergence, wavelength spread, incidence-angle uncertainty, calibrated detector mapping, and finite integration width all broaden the effective scattering condition. The projected coordinate should therefore not be interpreted as an exact cut through the reciprocal space of one perfect crystallite. The comparison remains direct because the same coordinate transformation and finite integration operation are applied to the calculated and measured detector quantities.

The off-specular and $m = 0$ profiles play complementary roles. The off-specular arcs test the two-dimensional-powder reciprocal-space construction and calibrated detector geometry. Their widths and asymmetries constrain the effective narrow mosaic component together with beam divergence and finite integration width. The off-condition $m = 0$ profiles are more sensitive to the broad component, but remain coupled to bandwidth, divergence, background, and alignment. The displayed overlays show that the fitted model can represent the selected feature locations and apparent profile shapes. A controlled Gaussian-only comparison is still required before the broad component can be called necessary or uniquely Lorentzian.

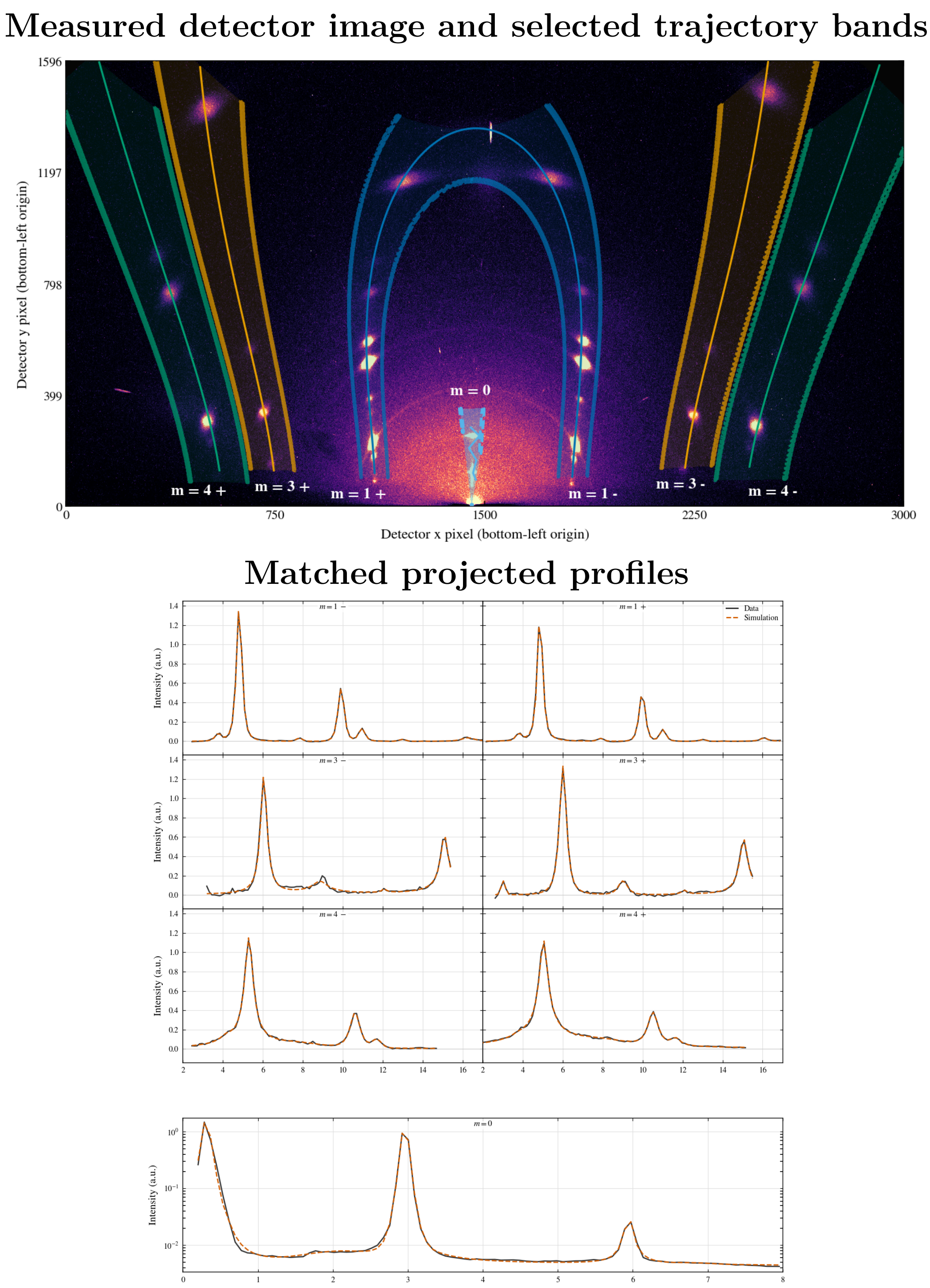

Figure 7: Bi_2Se_3 detector-space comparison at $\theta_i = 5°$. The upper panel is the measured detector image with calculated trajectory centerlines, finite integration regions, and their boundaries. Labels m identify radial rod families, while plus and minus signs distinguish the two detector-side intersections of one off-specular family. The lower panel compares measured black curves with calculated orange dashed curves generated using the same coordinate map, integration limits, and area normalization. The figure tests trajectory placement and apparent profile line shape. Separate family amplitudes are not a common-scale relative-intensity test, and the $m = 0$ row does not establish a unique broad-component functional form.

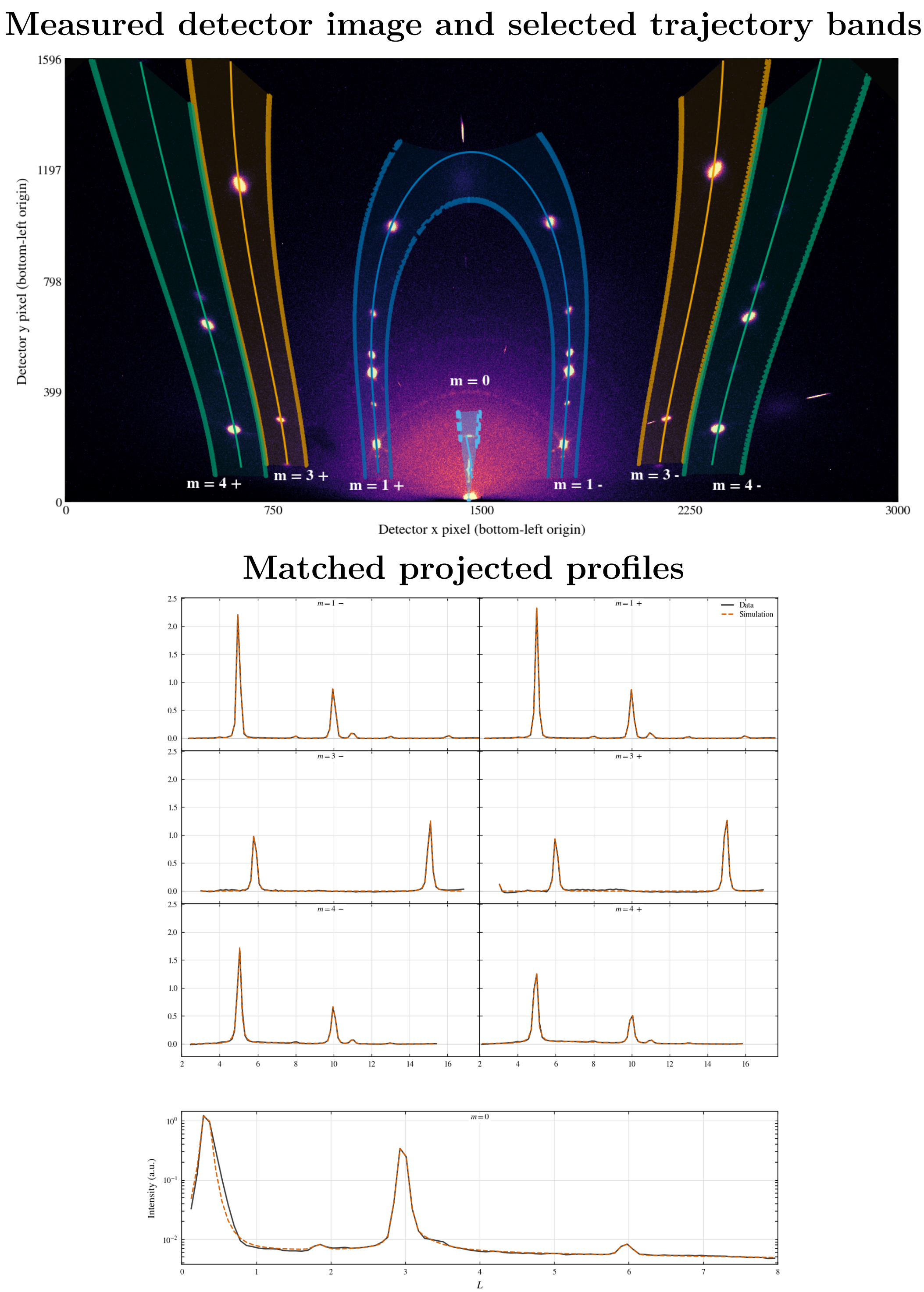

Figure 8: Bi_2Te_3 detector-space comparison at $\theta_i = 5°$, using the conventions of Fig. 7. The upper panel shows the measured detector image with calculated trajectories and finite integration regions. The lower panel compares detector-derived profiles formed with the same coordinate transformation, finite integration, and area normalization. The principal peaks occur at the same projected locations and have similar apparent line shapes within the displayed fitted condition. Separate family amplitudes are not interpreted on one common physical scale, and the $m = 0$ profile is not proof of a unique broad mosaic component.

4 Refinement workflow

The refinement is staged rather than fully global from the beginning. Many parameters affect similar detector-space observables: detector distance shifts peak positions, sample alignment changes the apparent incident condition, mosaicity changes profile widths and off-condition intensity, and structure factors change relative peak heights. A simultaneous fit can therefore hide physical errors by compensating one parameter group with another. The staged workflow fixes the instrumental quantities first, then introduces sample orientation, mosaicity, ordered intensity, and finally stacking disorder where needed.

The first distinction is between instrumental parameters and sample parameters. Instrumental parameters describe the beam and detector and should be shared across samples collected under the same experimental configuration. Sample parameters describe the film: incidence condition, sample offset, mosaic distribution, structural intensity, and, for PbI_2, stacking disorder. The refinement sequence in Table 1 uses the detector-space observable most sensitive to each group before that group is used in the final image projection.

Table 1: Staged refinement workflow. Steps 1–5 are used for the ordered Bi_2Se_3 and Bi_2Te_3 test cases. Step 6 is the added PbI_2 extension for stacking-fault diffuse scattering. All sample images are dark-subtracted before the listed observables are formed. The final measured and calculated profiles use the same calibrated coordinate map, finite integration limits, and normalization.

Step	Stage	Data	Parameters refined	Primary information
1	Beam characterization	Direct-beam images at several detector distances	Beam widths w_x, w_z and divergences $\Delta\theta_x$, $\Delta\theta_z$	Incident phase space
2	Detector geometry calibration	Powder calibrant image	Detector distance D, detector tilts (β, γ), and beam center (x_0, y_0)	Detector mapping
3	Sample alignment and goniometer misalignment	Indexed reflection centers from sample images	Sample geometry (θ_i, z_B, z_S), in-plane rotation χ, goniometer misalignment (α, ψ)	Detector-space peak positions
4	Mosaicity/profile refinement	Selected local detector ROIs, off-condition specular regions, and local profile shapes	Narrow-core plus broad-component mosaic parameters	Center-aligned profile shapes, off-specular arcs, and off-condition specular intensity
5	Ordered-film comparison	Selected detector-space Bragg regions and Q_z projections from fixed-m trajectories	Image-wise nuisance scales and constrained ordered-intensity parameters	Selected Bragg-feature locations and apparent line shapes
6	PbI_2 diffuse extension	Fixed-Q_R diffuse regions, ordered-baseline residuals, and projected Q_z profiles	Stacking-fault probabilities or layer-pair correlation parameters	Diffuse intensity along Q_z and separation of ordered and disorder-derived scattering

The instrumental stages fix where an ideal reflection should appear before any sample-specific physics is refined. Direct-beam images define the incident phase space, including beam width and angular divergence. A powder calibrant fixes the detector mapping: distance, beam center, and detector tilts. These two stages are intentionally separated from sample fitting because an incorrect detector geometry can otherwise be absorbed into apparent sample tilt, mosaic broadening, or shifted reciprocal-space trajectories.

The alignment stage is the part of the workflow most similar to detector-space indexing. Indexed sample reflections are used to determine how the film sits in the beam, fixing the nominal incident angle, sample offsets, in-plane rotation, and goniometer misalignment well enough that calculated Bragg-rod trajectories can be overlaid on the detector. This establishes where allowed reflections should appear, analogous to detector-space indexing tools such as indexGIXS.[29] Only after this geometric step does it make sense to refine the mosaic distribution, because mosaicity should explain profile shape and off-condition intensity rather than compensate for a misplaced trajectory.

The mosaicity stage constrains the shared narrow-core plus broad-component angular distribution. Local off-specular profiles constrain the narrow core, while the specular-family profiles are sensitive to the broad component together with bandwidth, divergence, alignment, and background. The ordered-film comparison then applies the same calibrated coordinate map and finite profile integration to the measured and calculated detector quantities. The present main-text figures use this stage to compare feature locations and apparent profile shapes. Quantitative relative-intensity refinement requires a documented common normalization and retained direct arrays.

For PbI_2, the ordered workflow is not discarded. It provides the geometry, projection, and mosaic baseline needed before diffuse scattering can be interpreted. The additional stacking-disorder stage asks whether a physically defined layer-correlation model accounts for the remaining diffuse intensity along fixed-Q_R cylinders. This keeps the PbI_2 result tied to the same detector-space framework while making clear which extra parameters belong specifically to stacking faults.

Implementation details, including weighted source-state quadrature, analytic Ewald restriction, detector-coordinate integration, parameter-correlation plots, and propagated uncertainty from the 2θ–ϕ and Q transformations, are documented in the Supporting Information. The main-text role of the workflow is to make the logic of the refinement reproducible. Instrumental geometry is determined first, sample orientation second, mosaic and profile effects third, and ordered or diffuse intensity differences are interpreted as structural information only after those stages are fixed.

5 Results: diffuse scattering and stacking disorder in PbI_2

Figure 9 shows a CVD-grown PbI_2 film and a crop spanning the $m = 0$ and $m = 1$ Bragg rods. The $m = 1$ rod contains measurable intensity between the integer Bragg maxima, whereas the $m = 0$ rod is dominated by sharp peaks and finite-thickness oscillations. This rod-dependent redistribution of intensity is the experimental motivation for the stacking-disorder model. In a two-dimensional powder, random in-plane rotation maps each in-plane reflection family onto a reciprocal-space cylinder at fixed Q_R. Perfect stacking concentrates intensity at discrete Q_z values on a cylinder. Loss of stacking coherence redistributes part of that intensity along the same cylinder.

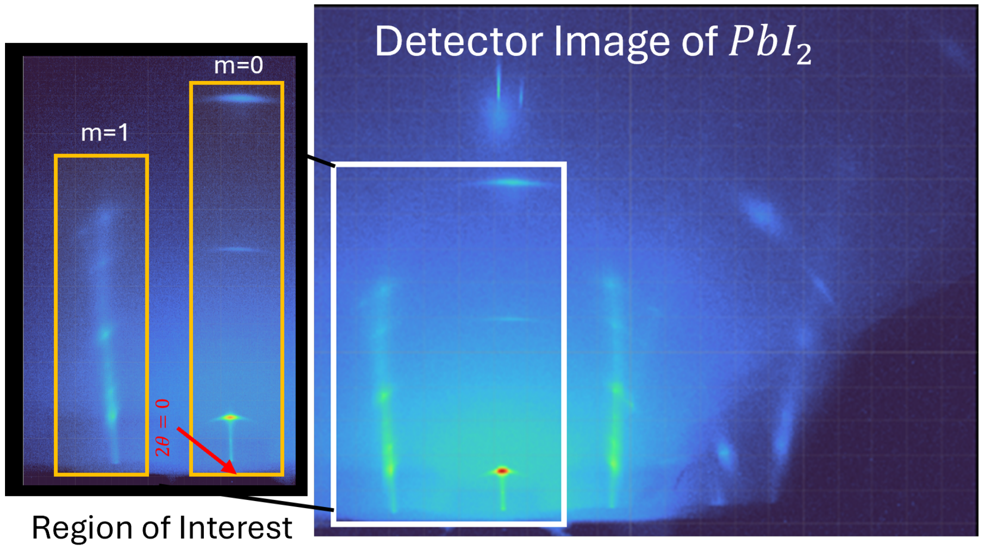

Figure 9: Measured PbI_2 diffraction image (right) and an enlarged region of interest (left). The direct-beam position defines the vertical $m = 0$ trajectory. Diffuse intensity is visible between the ordered maxima on the neighboring $m = 1$ rod, while the $m = 0$ profile remains insensitive to the lateral registry shifts that generate stacking contrast on other rods.

5.1 Direction-resolved transition-matrix model

The structural unit is one I–Pb–I trilayer. Its state is written $\alpha = (r, \sigma)$, where $r \in \{0, 1, 2\}$ denotes one of three basal registries and $\sigma \in \{+, -\}$ denotes the two trilayer orientations. The six states are therefore

$$\mathcal{S} = [0F_+, 1F_+, 2F_+, 0F_-, 1F_-, 2F_-]\,. \tag{11}$$

For a fixed m rod, a forward registry step $\mathbf{s} = (1/3, 2/3)$ contributes

$$\omega_{hk} = \exp[2\pi i(hs_x + ks_y)] = \exp\left[\frac{2\pi i}{3}(h + 2k)\right], \tag{12}$$

while the two local orientations carry form factors $F_+(h, k, L)$ and $F_-(h, k, L)$.

Adjacent trilayers obey the first-order interface law

$$\boldsymbol{\theta} = (a, b_+, b_-, d_+, d_-), \qquad a + b_+ + b_- + d_+ + d_- = 1. \tag{13}$$

Here a denotes no registry or orientation change. b_+ and b_- are forward and backward registry shifts without an orientation change. d_+ and d_- are the two handed shift-plus-orientation-change events. A change of orientation without a registry shift is excluded. The selected deterministic limits are $a = 1$ for 2H, $d_+ = 1$ for 4H, and $b_+ = 1$ for 6H. The opposite registry directions generate their symmetry-related handed counterparts. Figure 10 renders these cycles as projected I–Pb–I stacks. The sequences are read from bottom to top: $0F_+ \to 0F_+$ for 2H, $0F_+ \to 1F_- \to 0F_+$ for 4H, and $0F_+ \to 1F_+ \to 2F_+ \to 0F_+$ for 6H. These parent structures and PbI_2 polytypism are established experimentally.[32, 33]

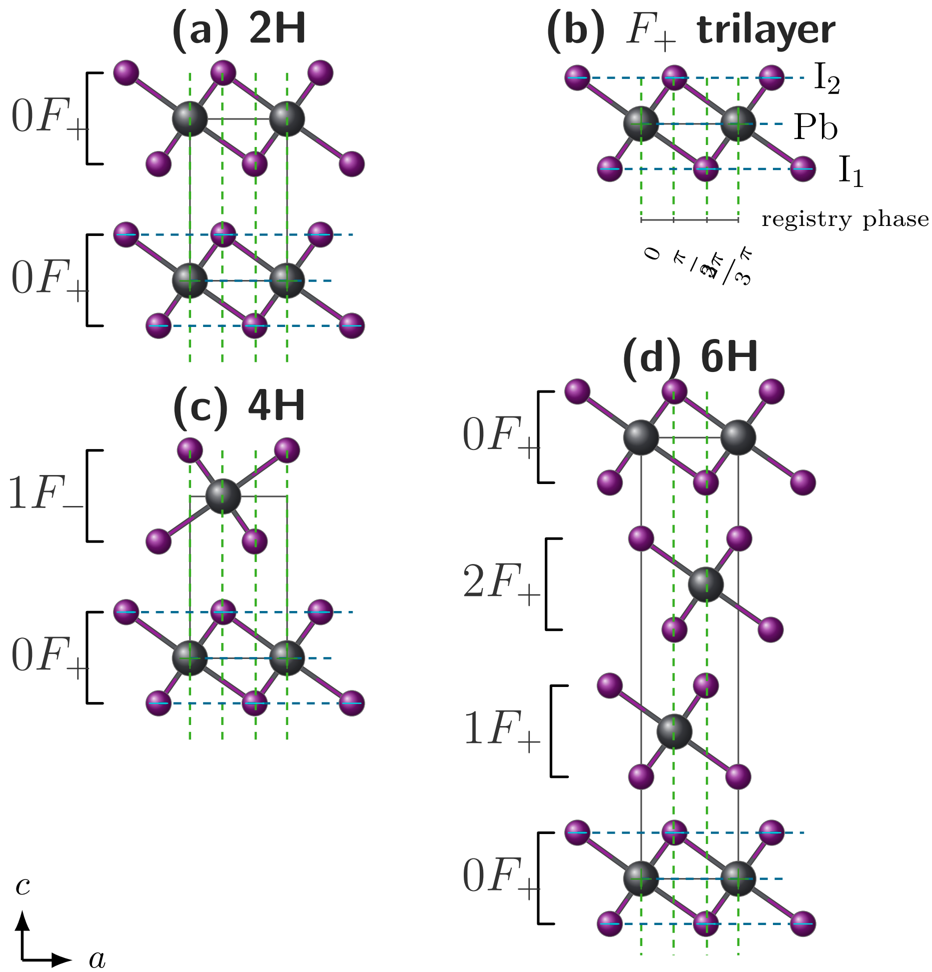

Figure 10: Projected I–Pb–I trilayers for the deterministic PbI_2 parent cycles used in the transition model. (a) 2H repeats $0F_+$ under $a = 1$. (b) An isolated F_+ trilayer shows the Pb and I atomic planes used to define the local orientation. (c) 4H alternates $0F_+$ and $1F_-$ under $d_+ = 1$. (d) 6H advances $0F_+ \to 1F_+ \to 2F_+ \to 0F_+$ under $b_+ = 1$. The c axis points upward, so each stack is read from bottom to top. In a state label rF_σ, $r \in \{0, 1, 2\}$ is the basal registry and $\sigma \in \{+, -\}$ is the trilayer orientation. Violet and gray spheres denote I and Pb, green vertical guides mark lateral registry, blue horizontal guides mark atomic planes, and brackets group each I–Pb–I trilayer. Signs are shown on the reference F_+ trilayer.

The general layer-correlation treatment follows the established transition-matrix theory of one-dimensionally disordered crystals, whereas the six states and allowed PbI_2 interface events above are the material-specific construction used here.[16, 34–36] Because the registry labels enter a given rod only through 1, ω_{hk}, and ω_{hk}^{-1}, the full six-state matrix reduces exactly to the following phase-weighted matrix in the orientation basis:

$$M(\omega) = \begin{pmatrix} a + b_+\omega + b_-\omega^{-1} & d_+\omega + d_-\omega^{-1} \\ d_+\omega^{-1} + d_-\omega & a + b_+\omega + b_-\omega^{-1} \end{pmatrix}. \tag{14}$$

The probability matrix $P = M(1)$ propagates the orientation populations, while $M(\omega)^\Delta$ sums the probabilities and registry phases of all paths connecting two trilayers separated by Δ interfaces. The complete construction and its reduction are given in Supplementary Note S3..

For each parent-rich population $j \in \{2H, 4H, 6H\}$, the disorder parameter has a direct probabilistic meaning:

$$p_j(\text{parent event}) = 1 - \epsilon_j, \qquad p_j(\text{each of the four alternative events}) = \frac{\epsilon_j}{4}. \tag{15}$$

For 2H, 4H, and 6H the parent event is a, d_+, and b_+, respectively. Written in the order (a, b_+, b_-, d_+, d_-), the five probabilities in Eq. (15) form $\boldsymbol{\theta}_j$ and are inserted directly into $M_j(\omega)$ and $P_j = M_j(1)$. Thus ϵ_j changes the matrix powers that determine how much phase coherence remains between increasingly distant trilayers.

For one definite sequence $s_n = (r_n, \sigma_n)$, the layer amplitude is $g_{s_n} = \omega^{r_n} F_{\sigma_n}$ and the coherent stack amplitude is $A_N(L) = \sum_{n=0}^{N-1} \xi(L)^n g_{s_n}$, where $\xi(L) = \exp(iq_z d)$ is the vertical phase between neighboring trilayer centers. Expanding $\langle |A_N|^2 \rangle$ leaves two familiar contributions: the intensity scattered by each trilayer individually and the interference between distinct trilayer pairs.

For population j, define the orientation population of trilayer n as

$$\boldsymbol{\pi}_{j,n} = \left(p_{j,n}^+, p_{j,n}^-\right) = \boldsymbol{\pi}_{j,0} P_j^n, \qquad p_{j,n}^+ + p_{j,n}^- = 1. \tag{16}$$

Here j identifies the 2H-, 4H-, or 6H-rich population and n identifies a trilayer within the finite stack. The quantities $p_{j,n}^+$ and $p_{j,n}^-$ are ordinary probabilities that trilayer n has orientation F_+ or F_-, respectively. They are neither scattering amplitudes nor the sample weights w_j used below.

The disorder-averaged interference amplitude for trilayers n and $n + \Delta$ is

$$K_{n,\Delta}^{(j)} = \begin{pmatrix} p_{j,n}^+ & p_{j,n}^- \end{pmatrix} \begin{pmatrix} F_+^* & 0 \\ 0 & F_-^* \end{pmatrix} M_j(\omega)^\Delta \begin{pmatrix} F_+ \\ F_- \end{pmatrix}. \tag{17}$$

The finite-stack intensity is then written as "self scattering plus pair interference":

$$I_j(L) = I_{j,\mathrm{self}}(L) + I_{j,\mathrm{pair}}(L), \tag{18a}$$

$$I_{j,\mathrm{self}}(L) = \frac{1}{N}\sum_{n=0}^{N-1}\left[p^{+}_{j,n}|F_+|^2 + p^{-}_{j,n}|F_-|^2\right], \tag{18b}$$

$$I_{j,\mathrm{pair}}(L) = \frac{2}{N}\,\mathrm{Re}\sum_{\Delta=1}^{N-1}\xi(L)^{\Delta}\sum_{n=0}^{N-1-\Delta}K^{(j)}_{n,\Delta}. \tag{18c}$$

The self term averages the isolated-trilayer intensity over the two possible orientations. The pair term groups all trilayer pairs by their separation Δ: there are $N-\Delta$ pairs at that separation, and each carries the same vertical phase $\xi(L)^{\Delta}$. In Eq. (17), the expression can be read from right to left. The final column supplies the scattering amplitude of the later trilayer, $M_j(\omega)^{\Delta}$ sums all allowed stacking paths and registry phases between the two layers, the diagonal matrix supplies the conjugated amplitude of the earlier trilayer, and $\boldsymbol{\pi}_{j,n}$ averages over whether that earlier trilayer is in the F_+ or F_- orientation. The factor $2\,\mathrm{Re}$ in Eq. (18c) includes the reverse-ordered conjugate pair without writing it separately. The complete derivation is given in Supplementary Note S3.

The 2H-, 4H-, and 6H-rich domains are taken to be mutually incoherent, so the three calculated intensities (not their amplitudes) are combined as

$$I_{\mathrm{calc}}(L) = S\left\{\left[w_{2H}I_{2H}(L) + w_{4H}I_{4H}(L) + w_{6H}I_{6H}(L)\right]\otimes K_{\mathrm{prof}}(L)\right\}, \qquad w_{2H}+w_{4H}+w_{6H} = 1. \tag{19}$$

Each $I_j(L)$ is first evaluated from Eq. (18) using its own ϵ_j. The weights then specify the fraction of the total calculated intensity assigned to each parent-rich population, the declared fixed profile kernel $K_{\mathrm{prof}}(L)$ represents the finite upstream profile integration and broadening used for the selected comparison, and S converts the normalized calculation to the measured intensity scale. Accordingly, ϵ_j is an effective per-interface disorder probability under the stated equal-alternative template, whereas w_j is an effective population weight.

For $h = k = 0$, $\omega_{hk} = 1$ and $F_+ = F_-$. Registry shifts and orientation changes then leave the scattering amplitude unchanged, so stacking faults are contrast-matched rather than absent. The oscillations on the $m = 0$ rod are therefore finite-thickness fringes. The same criterion applies to any other rod for which both the registry phase and the orientation form factors are indistinguishable.

5.2 Refinement

The disorder stage is applied only after the detector geometry, sample alignment, mosaic distribution, lattice constants, ordered PbI_2 structure factor, and fixed profile construction have been established by the upstream refinement. Measured and calculated detector images are projected through the same constant-Q_R rod masks and reported versus Q_z or the corresponding stacking coordinate L. No independent peak areas are introduced before the stacking model is evaluated, because doing so would absorb the diffuse between-peak intensity that the transition law is intended to explain.

The core disorder parameters are

$$\boldsymbol{\Theta}_{\mathrm{dis}} = (\epsilon_{2H}, \epsilon_{4H}, \epsilon_{6H}, w_{2H}, w_{4H}, w_{6H}, N), \qquad w_{2H} + w_{4H} + w_{6H} = 1, \tag{20}$$

so the three weights contain only two independent degrees of freedom. A stable refinement can first impose a shared disorder magnitude $\epsilon_{2H} = \epsilon_{4H} = \epsilon_{6H}$ and release the three values only when the retained fault-sensitive rods support phase-dependent broadening. The selected-profile residual is

$$r_j = \frac{I_j^{\mathrm{meas}} - I_j^{\mathrm{model}}(\boldsymbol{\Theta}_{\mathrm{dis}})}{\sigma_j}, \tag{21}$$

with any allowed global scale or low-order background terms declared separately. The fit is accepted only when it improves the selected diffuse profiles without degrading the ordered detector-space agreement established in the preceding stages.

Disorder Series Fitting

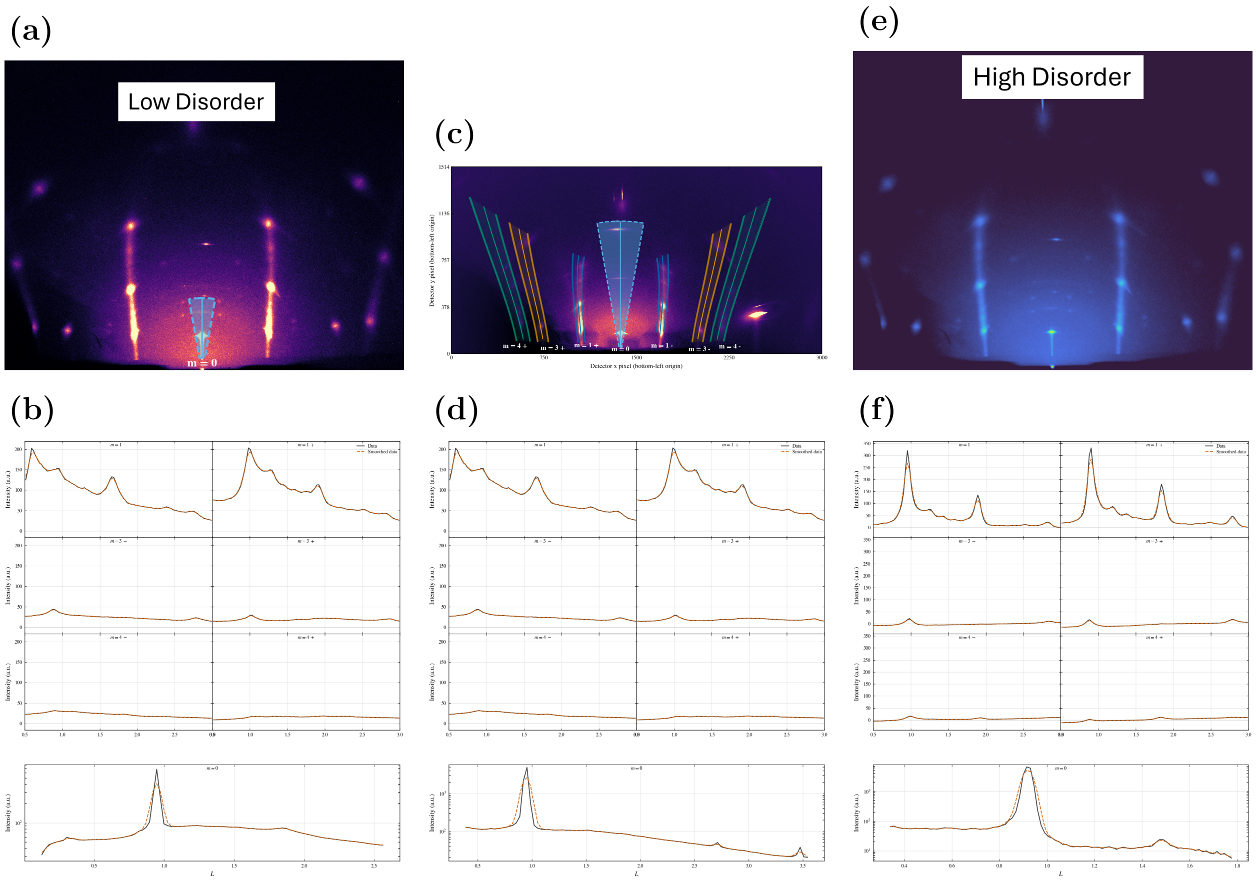

Figure 11: PbI_2 diffuse-rod comparison at $\theta_i = 4°$ for samples spanning increasing stacking disorder. The top row shows measured detector images and the constant-Q_R extraction regions. Solid curves mark the projected rod centerlines, shaded bands mark the integrated detector pixels, and dashed outlines mark the integration boundaries. The bottom row compares profiles obtained by applying the same projection to data and calculation. Black solid curves are measured and orange dashed curves are calculated. The series tests whether one detector-space geometry and ordered baseline, augmented by the transition-matrix stacking model, reproduce the progressive transfer of intensity from ordered maxima into the diffuse rod signal.

Figure 11 shows the visible progression from sharp ordered maxima to stronger between-peak intensity. The absence of comparable diffuse contrast on $m = 0$ does not imply that those scattering volumes contain no faults. It follows from the phase-insensitive condition described above. Likewise, changes in the visibility of higher order $m = 0$ peaks are assigned to the growth-dependent mosaic and profile-broadening state rather than to lateral-registry disorder.

The ordered baseline and the transition-model parameterization are summarized in Table 2. Numerical values from the superseded scalar two-population fit are not carried into the new model because they do not map uniquely onto three population-specific ϵ_j values and three normalized weights.

Table 2: PbI_2 ordered-baseline parameters and stacking-disorder parameterization for the selected-rod comparison. Numerical transition-matrix parameters must be reported from a refit using Eqs. (15)–(19). Values from the superseded scalar model are not reused.

Parameter class	Fitted value or model quantity	Literature comparison	Role in the selected-rod model
Lattice and coordinates	$a = 4.5583$ Å and $c = 6.985$ Å. Pb is at $(0, 0, 0)$. I1 is at $(1/3, 2/3, 0.2677)$, and I2 is at $(2/3, 1/3, 0.7323)$.	Lattice constants remain within the 2H single-crystal structural spread, and the iodine coordinate follows the corrected $z \simeq 0.268$ value.[32, 33]	Fixes the ordered Bragg positions and the $\|F_{hk}(L)\|^2$ envelope before stacking disorder is introduced.
Occupancies	A weakly floating test returned $o_{\rm Pb} = 0.999$, $o_{\rm I1} = 1.001$, and $o_{\rm I2} = 1.000$. The ordered baseline was then evaluated with full occupancies.	The clean ordered model uses full occupancies. The density-deficient occupancy reported for a Palosz-type sample is sample-specific and was not used as the default model.[33, 37]	Keeps the PbI_2 average structure stoichiometric while the diffuse rod intensity is assigned to stacking correlations.
Displacement factor	The current ordered-film interface uses the crystallographic isotropic baseline and, when released, one shared uniaxial pair U_R and U_z in the film frame. Numerical values are not reported for the pending transition-matrix refit.	General site-resolved anisotropic CIF tensors are not transferred into the present calculation. Any site-specific directional refinement would require an explicitly implemented and documented structural model.[38]	Moderates the ordered intensity envelope without implying unsupported site-resolved displacement parameters.
Transition-matrix stacking disorder	ϵ_{2H}, ϵ_{4H}, ϵ_{6H}, normalized weights w_{2H}, w_{4H}, w_{6H}, and finite stack length N. Final numerical values are pending the transition-matrix refit.	These are effective selected-rod descriptors under the stated parent rules and equal-alternative fault templates, not universal PbI_2 constants.	Controls the redistribution of the ordered rod envelope into broadened maxima, finite-thickness fringes, and diffuse between-peak intensity.

The physical interpretation is therefore deliberately conditional on the model. Each ϵ_j measures the probability of departing from the selected parent event within population j, the w_j describe the incoherent population mixture, and N controls the finite-stack interference. Numerical values should be inserted only after the experimental profiles have been refit with this parameterization.

6 Discussion and conclusion

The ordered Bi_2Se_3 and Bi_2Te_3 test cases compare measured and calculated detector features after the calibrated geometry, weighted incident beam phase space, sample alignment, continuous mosaic distribution, ordered structure weighting, and finite detector integration are applied. The same coordinate map and finite profile operation are used for measured and calculated detector quantities. Within the fitted data set, the selected profiles agree in their principal peak positions and apparent line shapes, including features that would be difficult to interpret from a conventional one-dimensional reduction alone.

Keeping the comparison in detector space is important because the constraining intensity is distributed among off-specular arcs, cap-like $m = 0$ features, weak off-condition specular peaks, and bright near-critical intensity. A conventional one-dimensional reduction can merge or remove these distinctions. Matched projection makes the data and calculation directly comparable, but it does not remove correlations among mosaicity, bandwidth, divergence, alignment, and background. The off-condition $m = 0$ features motivate testing a weak broad orientation component. The displayed comparisons do not establish that this component is uniquely Lorentzian or necessary relative to a fairly re-optimized Gaussian-only model.

This distinction also defines the relationship to detector-space indexing software. Programs such as indexGIXS calculate and overlay allowed reflection positions and are the relevant prior art for the geometric indexing stage.[29] The contribution here is the subsequent continuous detector-coordinate intensity and profile calculation. The same geometry is combined with the incident beam phase space, continuous orientation density, structure-factor weighting, and finite detector integration to compare measured and calculated feature locations and apparent profile shapes. Standard planar-interface refraction and attenuation corrections are included where required, but the optical treatment is not presented as a new contribution.

The PbI_2 section shows how the same detector-space framework can be extended from ordered Bragg-profile comparisons to diffuse scattering from stacking disorder. In that extension, detector geometry, beam phase space, mosaicity, and the profile construction remain fixed, while the reciprocal-space intensity is generated from a stacking-correlation model rather than a perfectly periodic ordered stack. Diffuse intensity should therefore be interpreted as stacking information only after geometry and mosaicity have been constrained by the ordered-film workflow.

The staged workflow is part of the scientific argument. Instrumental parameters are determined first, sample alignment second, mosaic and profile parameters third, and ordered or diffuse intensity parameters last. This ordering reduces parameter compensation and makes the final data and model comparison interpretable. The manuscript follows the same logic by showing the observation, identifying the physical inference, introducing only the model ingredient needed for that inference, and then comparing measured and calculated quantities on matched detector-derived axes.

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